



University
of Regina

Go far, *Together.*

ENSE 375 – Software Testing and Validation

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- Proof read the text for typing and grammar mistakes.

- Follow the IEEE Bibliography style for the references by selecting "References/ Citations & Bibliography/ Style".

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1 Introduction

- Give a brief description of the design and a summary of the relevant background information related to the topic. Give a rationale about what is needed and why.
- Give the reader an overview of what is in the next sections.
- Do not put any detailed results of your work here.

2 Design Problem

This section has the following two subsections:

2.1 Problem Definition

For this design project, our group defined the problem to be the process of planning and designing, then creating and thoroughly testing a Model View Controller website using proper industry standard testing practices covered in class. Constraints also apply, which we must consider and follow several of them; these include: Economic factors, Regulatory compliance, Reliability, Sustainability and Environmental Factors, Ethics, and Societal Impacts. Due to the nature of the project, the focus of the project is to gain practical experience using several different testing and validation techniques, including Boundary value testing, Equivalence class testing, Decision tables testing, State transition testing, and Use case testing. As well, two functions must be tested; using Path testing and Data flow. During this process, we will follow the structured engineering design process; such as considering requirements, research, and brainstorming. Including analyzing and iterating on potential functions and prototypes. One important idea we are considering is whether to have accounts, which has a large upside, and some downside to implementing, and our process will prototype and test it. Our approved topic/idea for this project is Group Leader, a scheduling app for groups. Each member adds their schedule and availability, then the program will find common times between them, allowing the group to all pick a preferred time. This has a lot of use cases, including just for friend groups to meet, or even in workplaces, when they would need to schedule a meeting. Our first objective is societal impacts, which is to create this website, which will allow users to easily schedule times for meetings between busy schedules. It should be intuitive, quick, and user should have an easy time inputting their schedules. We also aim to meet regulatory compliance and user security risks, by either finding a solution for a secure log in, or avoiding storing schedule information. For reliability, we aim to take advantage of the testing cycles to ensure reliable and stable user experience. For economic factors, we are hoping to complete the project with spending no money, and if databases are required we will try to host locally, the product will also be free. Delegating work is another area for this project, where we will track which group member contributes individually. During the development cycle, we will document this process thoroughly, including test cases we use, commenting code, and having a public github.

2.2 Design Requirements

This section has the following three subsections:

2.2.1 Functions

- Provide functions of the design project. Remember that the functions contain verbs.

2.2.2 Objectives

- Provide objectives of the design project. Remember that the objectives are specified as adjectives.

2.2.3 Constraints

- Provide constraints here. Remember that the constraints are binary (either satisfied or not).

3 Solution

In this section, you will provide an account of some solutions your team brainstormed to implement and test the project. Some solutions might not have all the desired features, some might not satisfy the constraints, or both. These solutions come up in your mind while you brainstorm ways of implementing all the features while meeting the constraints. Towards, the end you select a solution that you think has all the features, testable and satisfies all the constraints. Remember that an engineering design is iterative in nature!

3.1 Solution 1

Write a brief description of your first solution and provide the reasons in terms of testing for not selecting this one.

3.2 Solution 2

This is an improved solution but might not be the final solution that you select. Give a brief description of this solution here. Again focus on its testing attributes.

3.3 Final Solution

This is the final solution. **Explain why it is better than other solutions** (focus more on testing). You may use a table for comparison purposes. After providing the reason for selecting this solution, detail it below.

3.3.1 Components

What components you used in the solution? What is the main purpose of using individual component? What testing method did you employ for each component? Provide a block diagram (with a numbered caption, such as Fig. 1) representing the connectivity and interaction between all the components.

3.3.2 Environmental, Societal, Safety, and Economic Considerations

Explain how your engineering design took into account environmental, societal, economic and other constraints into consideration. It may include how your design has positive contributions to the environment and society? What type of economic decisions you made? How did you make sure that the design is reliable and safe to use?

3.3.3 Test Cases and results

What test suits did you design to test your prototype? How did you execute the test cases to test the prototype?

3.3.4 Limitations

Every product has some limitations, and so is the case with your design product. Highlight some of the limitations of your solution here.

4 Team Work

Since this is a group project, you must have a fair distribution of tasks among yourselves. To this end, you must hold meetings to discuss the distribution of tasks and to keep a track of the project progress.

4.1 Meeting 1

Time: September 17 2026, around 1 pm.

Agenda: Distribution of Project Tasks

Team Member	Previous Task	Completion State	Next Task
James Clinton	N/A	Project Definition, Github creation, Project File Template	2.2.2 Objectives
Peyton Bowen	N/A	Business Case	2.2.1 Functions
Jaydon Forrester	N/A	Cost/Benefit	2.2.3 Constraints

4.2 Meeting 2

Time: Month Date, Year, hour: minutes am/pm to hour: minutes am/pm

Agenda: Review of Individual Progress

Team Member	Previous Task	Completion State	Next Task
Team member 1	Task 1	80%	Task 1, Task 5
Team member 2	Task 2	50%	Task 2
Team member 3	Task 3	100%	Task 6

4.3 Meeting 3

Provide a similar description here.

4.4 Meeting 4

Provide a similar description here.

5 Project Management

c denotes critical path									
Task	Slack Time LSD – ESD	Earliest Start Date	Latest Start Date	Start (DD/MM)	End (DD/MM)	Week 1	Week 2	Week 3	
First Deliverable									
Problem Definition	0 Days	17/09/26 1pm,	17/09/26 1pm	16/09/26	17/09/26	c			
Business Case	0 Days	17/09/26 1pm,	17/09/26 1pm	17//09/26	17//09/26	c			
Report MD	0 Days 9 hours	17/09/26 1pm,	17/09/26 10pm	17//09/26	17//09/26				
Project File Template	0 Days 9 hours	17/09/26 1pm,	15/09/26 10pm	17//09/26	17//09/26				
Second Deliverable									

6 Conclusion and Future Work

- A summary of what you achieved. Mention all the design functions and objectives that you achieved while satisfying testing requirements?
- While keeping the limitations of your solution, provide recommendations for future design improvements.

7 References

- Use the IEEE reference style.
- Do not put any reference if it is not cited in the text.

8 Appendix

If you want to provide an additional information, use this appendix.